18. VASCULAR SYSTEM : BRANCHIAL ARCHES

DURATION : 08:04
STUDY SHEET

COURSE SUMMARY

This video explores the development of the circulatory system from blood vascular islands, focusing on the evolution of the primitive aortic system and cardinal veins. The pharyngeal arches, which play a fundamental role in vascular and skeletal structure, are analysed in detail, including their derivatives and their influence on the skull and digestive tract. In addition to exploring vascular relationships, the video highlights the importance of balance in blood and lymphatic circulation, offering practical insights for osteopathy. The specific functions of the pharyngeal arches and their morphological contributions to structures such as the maxillary artery and the hyoid system are also discussed, providing a deep understanding necessary for effective therapeutic manipulations.

VASCULAR SYSTEM: BRANCHIAL ARCHES

We will explore the origin and development of the **circulatory system**, which begins within the blood vascular islands, located in the duplication of the **vitelline vesicle**. This process accelerates with the development of the **amniotic cavity** and the progressive integration of this vascular system.

THE PRIMITIVE AORTIC SYSTEM AND CARDINAL VEINS

We observe the appearance of a system called the **primitive aortic system**, including the future developing heart, as well as the **cardinal veins**. The latter, also called **mesonephric veins**, play a very important purifying role in connection with the first forming kidneys. The integration of these vessels into the aortic system becomes increasingly pronounced.

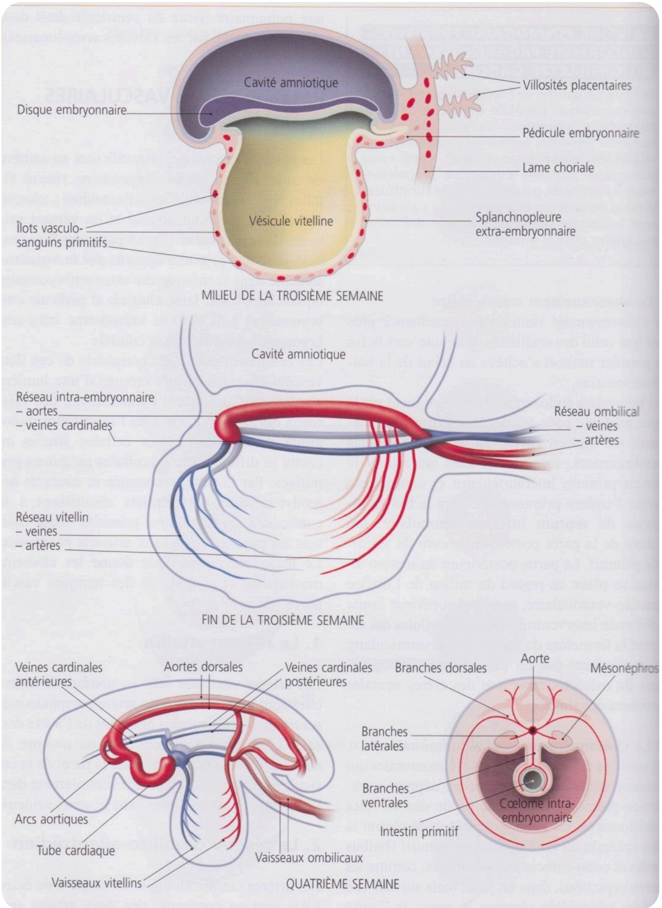

FIG. — Embryonic vascular development

IMPORTANCE OF VITELLINE AND UMBILICAL VEINS

The **vitelline veins** will initially form a plexus called the **vitelline artery plexus**. Its remnant will later become the **mesenteric system**. We also see the **umbilical arteries**, which will later join.

THE BRANCHIAL ARCHES

Our digestive tract is adjacent to structures called **branchial arches**. There are six of them, from pharyngeal pouch number 1 to 6. It is important to note that arch number 5 appears and disappears very quickly, which is why it is sometimes omitted in some descriptions, but it is indeed initially present.

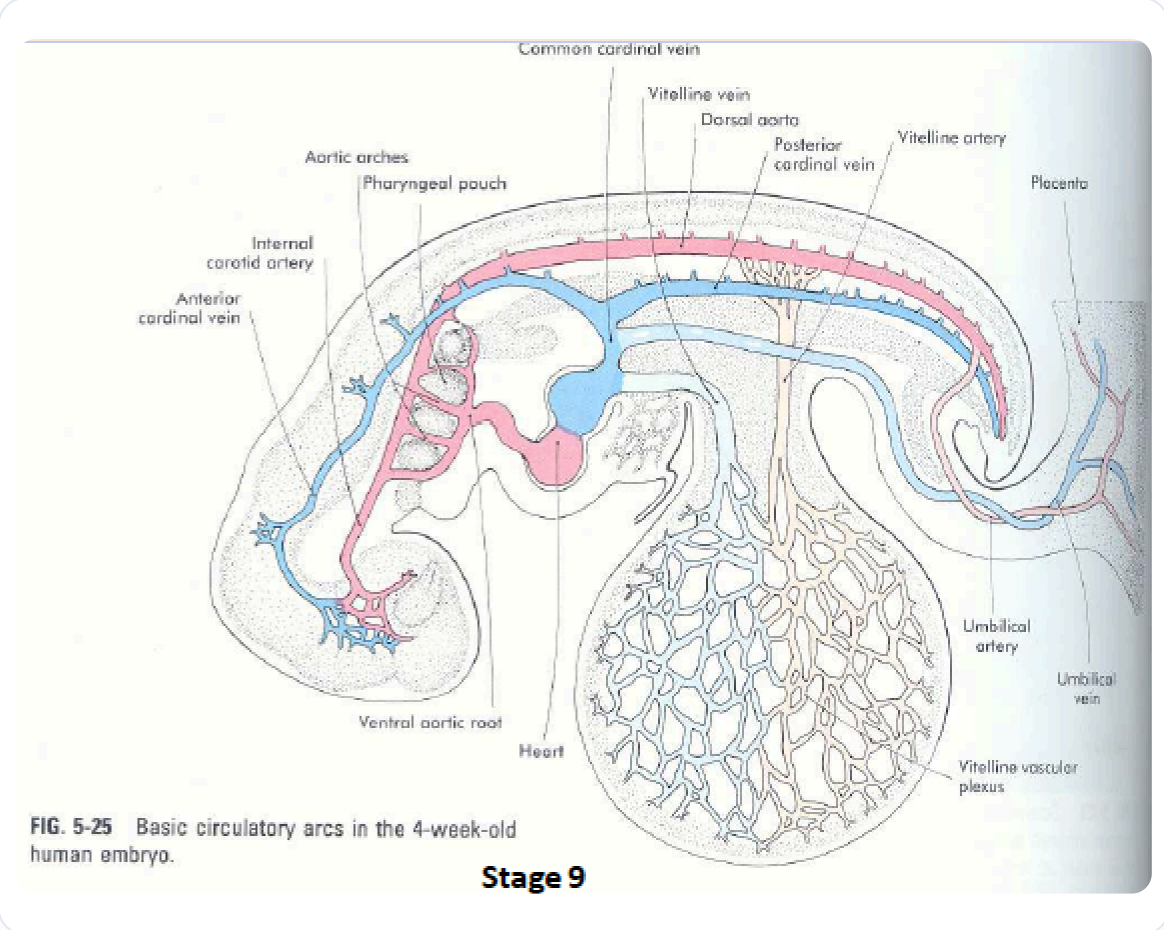

FIG. — Embryo circulation 4 weeks

VASCULAR EVOLUTION AND THE ROLE OF THE CRANIUM

It is fascinating to observe the transformations of this vascular system. One of the key ideas to remember is that the **aortic arch** derives from multiple segments. When working on the cranium, in relation to the brain and its vascular system, it is crucial to understand that the brain is primarily vascularised by the **carotid arteries** and the **posterior vertebral arteries**. These four vascular entry points represent essential areas to check and work on.

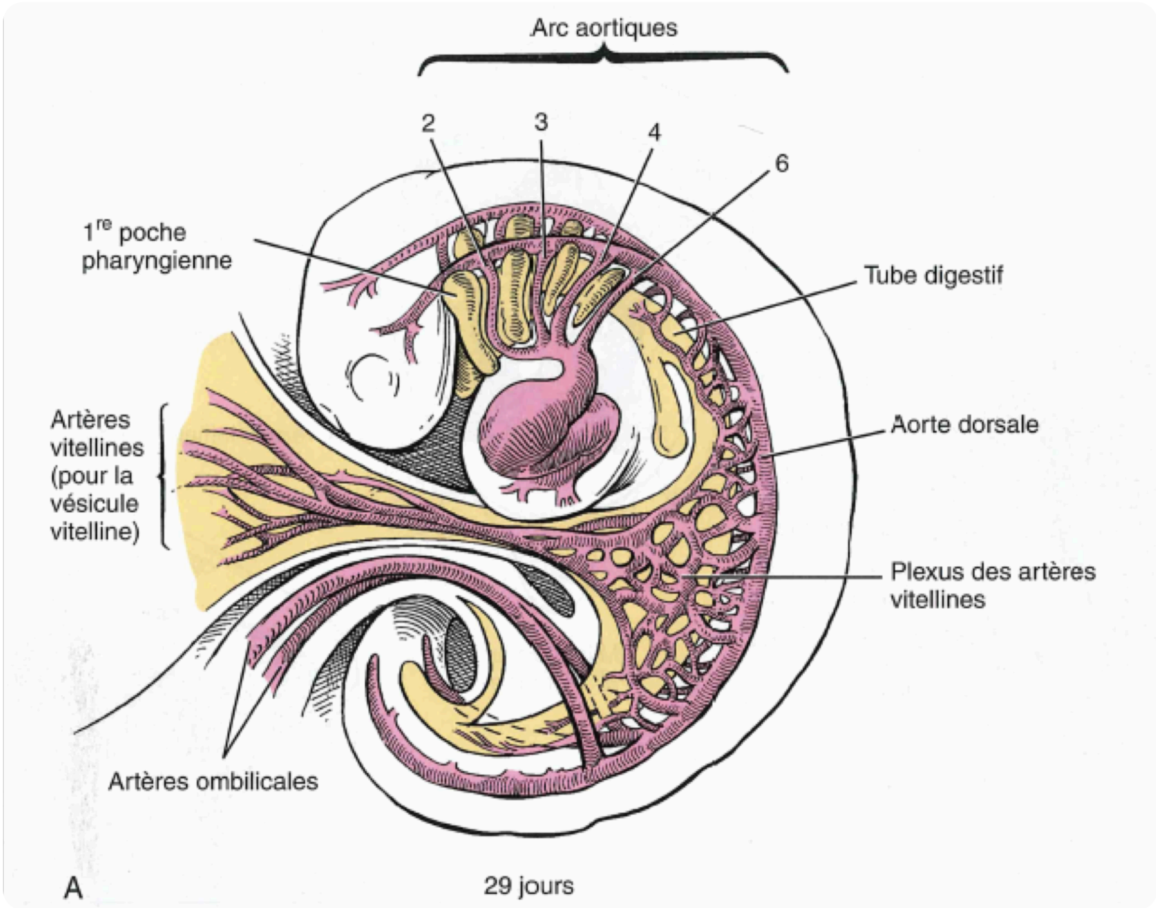

FIG. — Embryo 29 days

BALANCE AND ENERGY FLOWS

Circulation is a matter of **balance** at the capillary beds. This is where the exchange between the arterial system and the venous system occurs. These areas are places of energy transfer, where the balance between what enters and what leaves is paramount.

FORMATION OF THE LYMPHATIC SYSTEM

The arterial and venous system, combined with the resultant at the capillary beds, gives rise to the **lymphatic system**. This is a "transpiration" that creates a specific cleansing network. It is therefore essential to have areas for irrigation and for decongestion, in continuity with the **vertebral plexus**.

SPECIFIC ARTERIAL STRUCTURES OF THE BRANCHIAL ARCHES

Each branchial arch gives rise to a specific arterial structure.

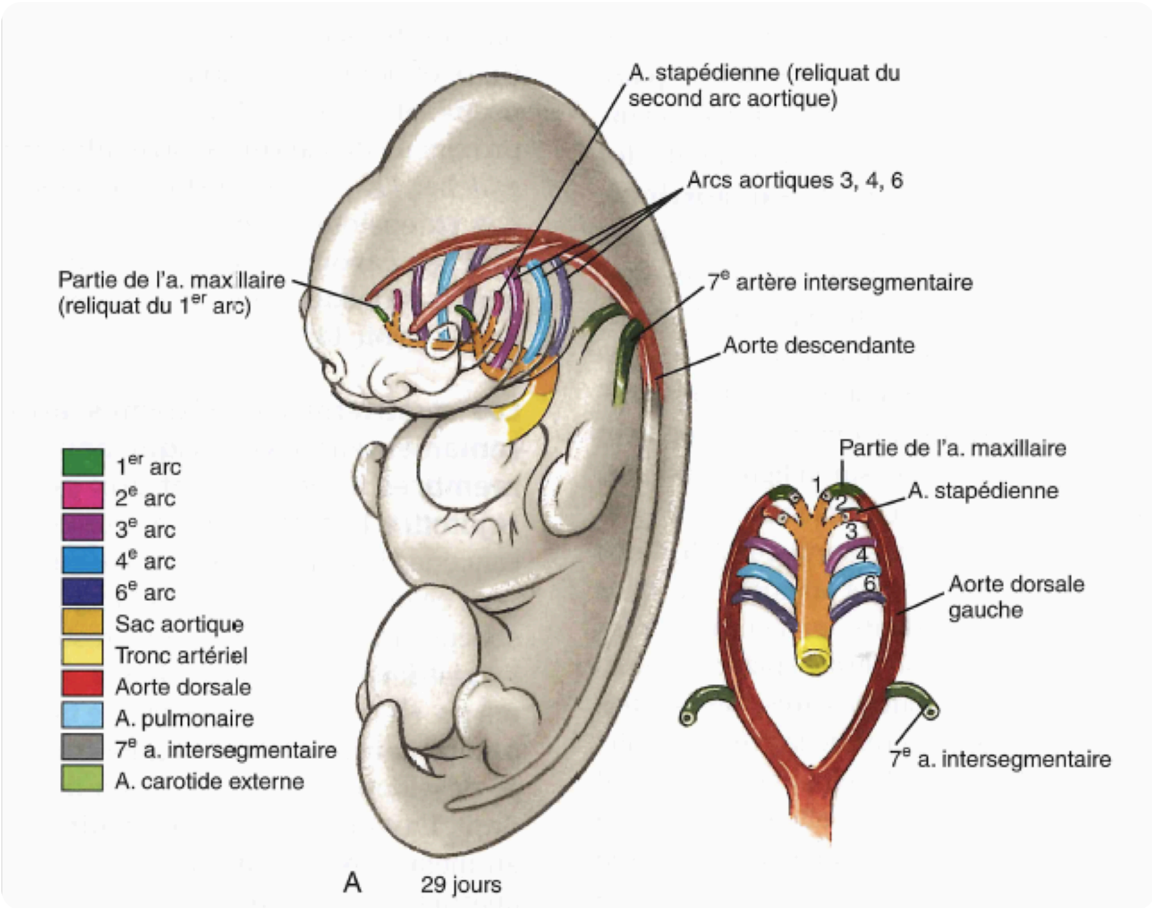

FIG. — Embryonic aortic arches

BRANCHIAL ARCH NUMBER 1

The remnant of branchial arch number 1 will form the **maxillary artery**. Skeletally, its cartilage will give rise to all derived structures up to the inner ears. For children, this area is particularly sensitive. They generally do not like their cranium to be touched, but are more receptive to manipulations of the viscerocranium.

CRANIUM-DIGESTIVE TRACT RELATIONSHIP

The brain forms the cranium, while the surrounding space influences the formation of the digestive tract. Remember the lines of force, or **dural sheaths** (anterior, middle, posterior), which orient towards the base of the cranium. All branchial arches also converge towards this base, which acts as an essential reference and support point (vascular, neurological, visceral). It is therefore important to balance these posterior, middle, and anterior parts.

DETAILS OF THE BRANCHIAL ARCHES AND THEIR DERIVATIVES

Here is an overview of the vascular and cartilaginous derivatives of the branchial arches:

- **Branchial arch number 1:**

* Maxillary artery.

* Meckel's cartilage, important for the stabilisation of the tympanic conduit.

* It is essential to reconnect this movement and rebalance it towards its fulcrum, the base of the cranium, allowing space to operate.

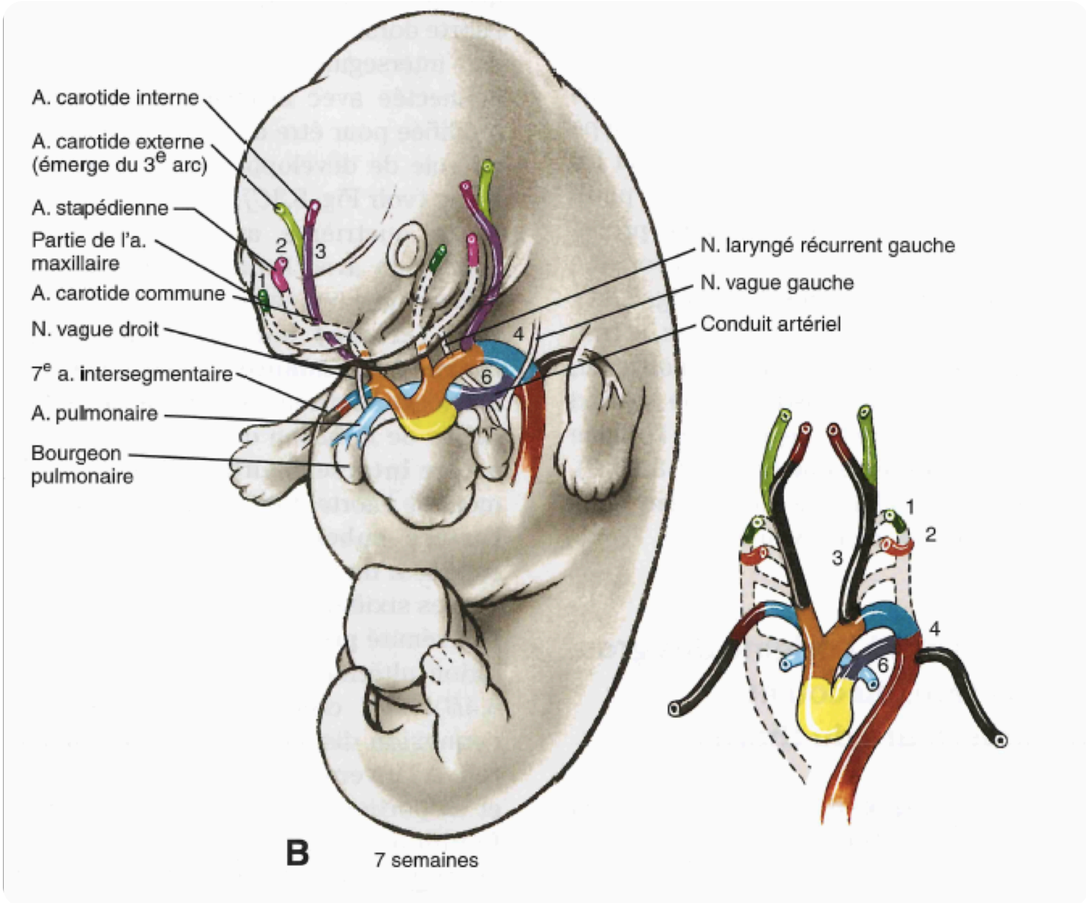

FIG. — Embryonic vascular system

- **Branchial arch number 2:**

* The hyoid system, from the stapedial and tympanic artery (which gives the stapedial artery), to the cartilages of the superior ring just at the thymic level.

* This system organises itself vascularly in a coherent manner.

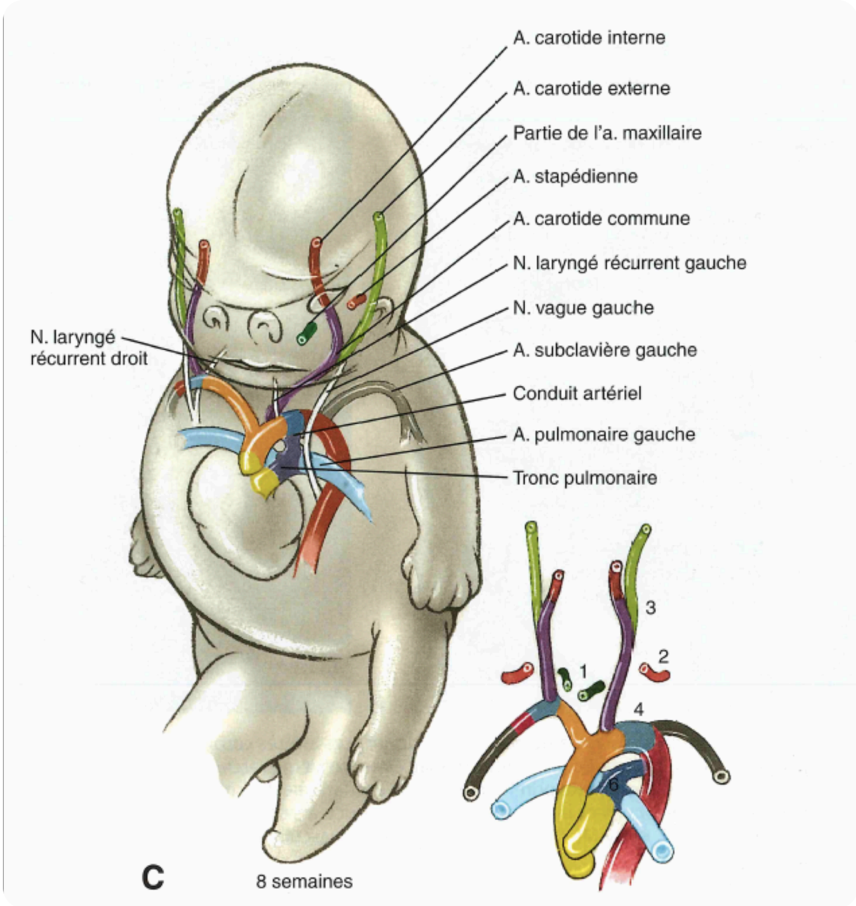

FIG. — Embryonic aortic arches

- **Branchial arch number 3:**

* The common carotid artery.

- **Branchial arch number 4:**

* The aorta and subclavian arteries.

- **Branchial arch number 5:**

* There is no persistent arterial derivative.

- **Branchial arch number 6:**

* The pulmonary artery.

Tableau 12 - 1. Dérivés des arcs pharyngiens et leurs tissus d'origine

ARC PHARYNGIEN	ARTERE DE L'ARC[a]	ELEMENTS SQUELETTIQUES	MUSCLES	NERF CRANIEN[b]
1	Branche terminale de l'artère maxillaire	Dérivés des cartilages (provenant de la crête neurale): du cartilage maxillaire: alisphénoïde, enclume cartilage mandibulaire (Meckel): marteau Dérivés par ossification directe du mésenchyme dermique de l'arc: maxillaire, zygomatique, portion squameuse de l'os temporal, mandibule	Muscles de la mastication (temporal, masséter et ptérygoïdiens), mylo-hyoïdien,ventre antérieur du digastrique, tenseur du tympan, tenseur du voile du palais (provenant du somitomère crânien 4)	Divisions mandibulaire et maxillaire du nerf trijumeau (V)
2	Artère stapédienne (de l'embryon), artère carotico-tympanique (de l'adulte)	Etrier, processus styloïde, ligament stylo-hyoïdien, petites cornes et bord supérieur de l'os hyoïde(dérivés du cartilage du second arc [Reichert]; provenant de la crête neurale)	Muscles de la mimique (orbiculaire de l'oeil, orbiculaire de la bouche, risorius, platysma, auriculaire, fronto-occipital et buccinateur), ventre postérieur du digastrique, stylo-hyoïdien, de l'étrier (provenant du somitomère crânien 6)	Nerf facial (VII)
3	Artère carotide commune, racine de l'artère carotide interne	Bord inférieur et grandes cornes de l'os hyoïde (provenant du cartilage du troisième arc;) issu de la crête neurale)	Stylo-pharyngien (provenant du somitomère crânien 7)	Nerf glosso-pharyngien (IX)
4	Arc de l'aorte, artère subclavière droite; bourgeons initiaux des artères pulmonaires	Cartilages du larynx (dérivés du cartilage du quatrième arc provenant du mésoblaste de la lame latérale)	Constricteurs du du pharynx, crico-thyroïdien, élévateur du voile du palais (issus des somites occipitaux 2 à 4)	Branche laryngée supérieure du nerf vague (X)
6	Conduit artériel, bourgeons des artères pulmonaires définitives	Cartilages du larynx (dérivés du cartilage du sixième arc; issus du mésoblaste de la lame latérale)	Muscles intrinsèques du larynx (issus des somites occipitaux 1 et 2)	Branche laryngée récurrente du nerf vague (X)

FIG. — Pharyngeal arch derivatives

All these systems form a functional unit, including, for example, the **laryngeal cartilages**, the **intracellular muscles**, and the **recurrent branch of the vagus nerve**. We work on these units by offering points of support, allowing the lines of motility to develop and the neutrals to meet, letting space contain the primary respiration.

THE FACE AND THE NEURAL CREST

The face has enormous **vascularisation**. Your facial bones, your skin, your facial cartilages are all derived from the **neural crest**, from the **mesoectoderm**. In addition to the points of support, we observe what is called the **neural crest stream**: prosencephalon, mesencephalon, rhombencephalon, followed by rhombomeres 1 to 7. This stream, this invasion of the neural crest into the mesoectoderm, is crucial developmental information.

ORGANISATION AROUND THE BASE OF THE CRANIUM

All the organisation of the branchial arches and the lines of force will tend towards this fundamental point of support which is the **base of the cranium**, corresponding to the tip of the **notochord**.

ROLES OF THE CIRCULATORY SYSTEMS

- **Arterial system:** It generates movement and ensures a homeostatic balance of volume and pressure.
- **Venous system:** It is a level of exchange, more osmotic.
- **Lymph:** It is responsible for elimination, particularly toxic elimination, and plays a role in immunity.

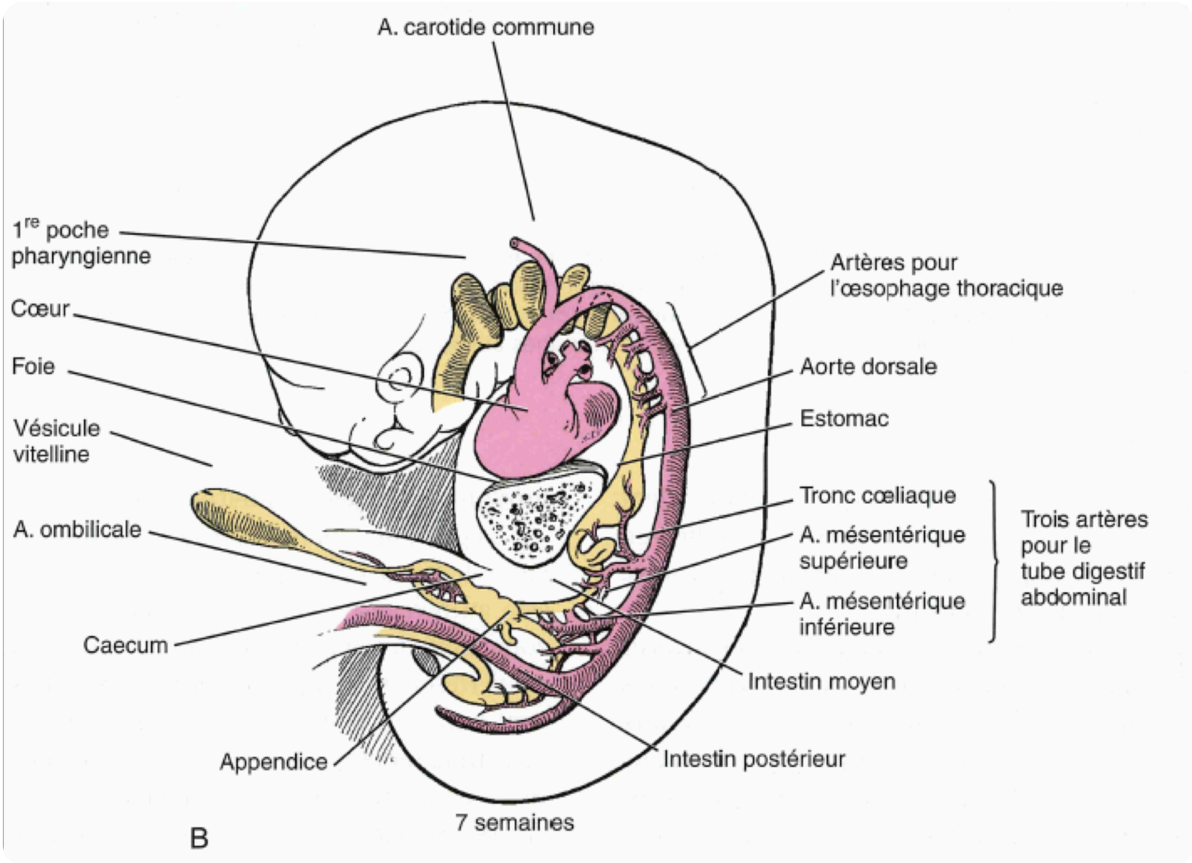

FIG. — Embryo 7 weeks, digestive system

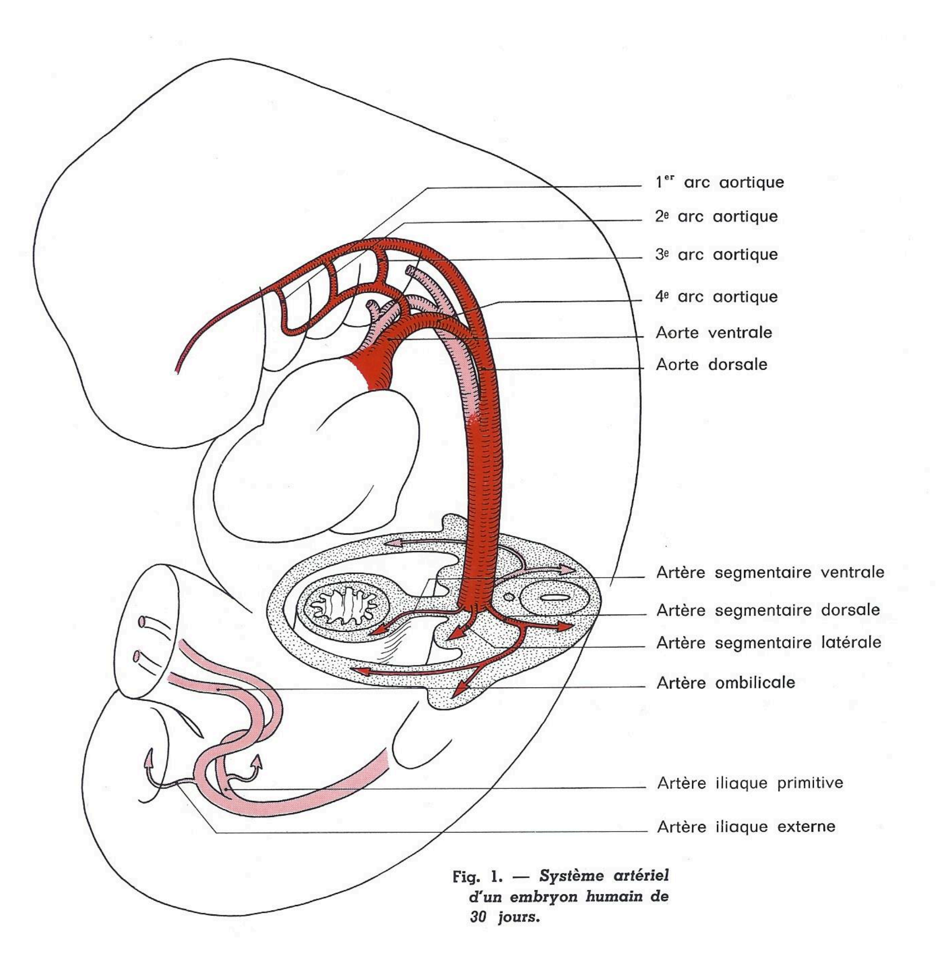

FIG. — Embryonic arterial system

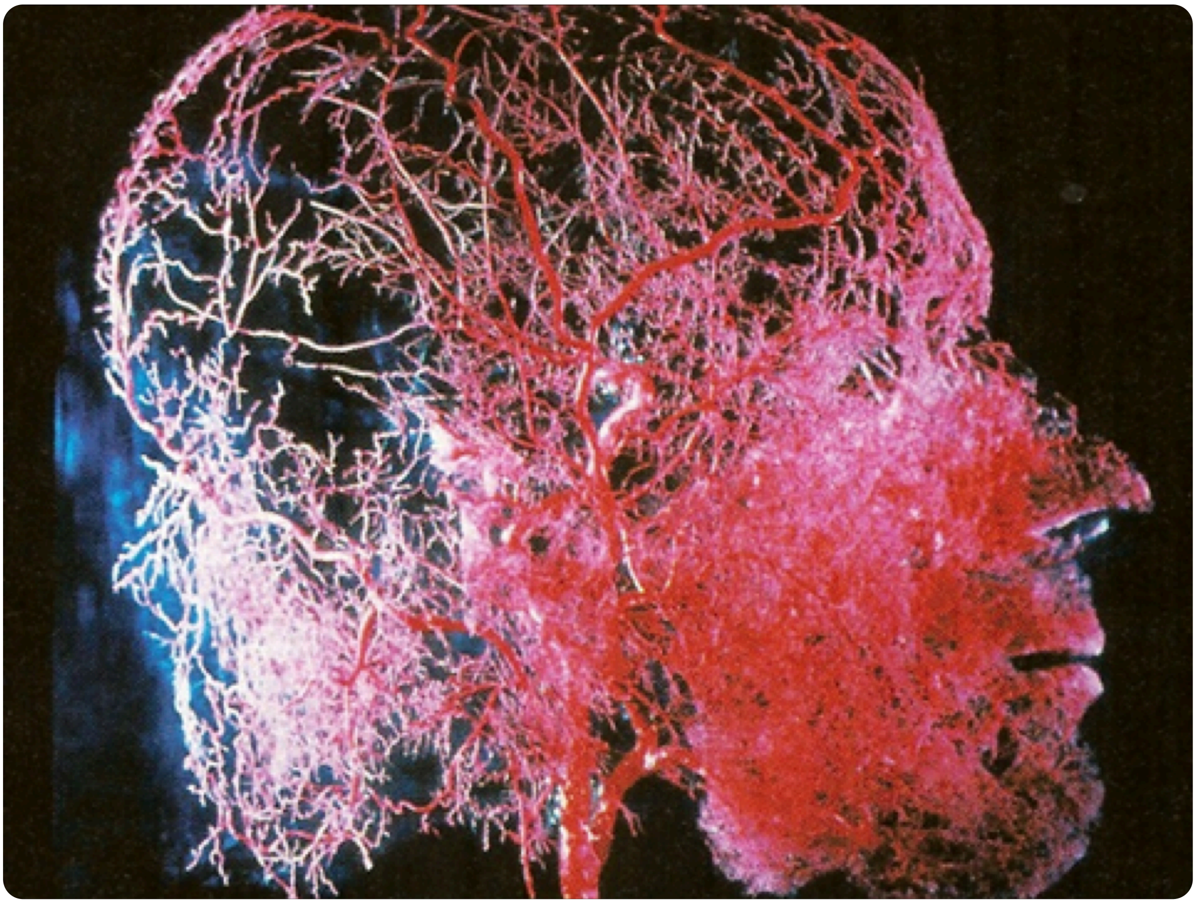

FIG. — Human head vascularization

The **cerebrospinal fluid** (CSF) is the controller. It ensures the specific molecular and hormonal balance of the body. Its return occurs according to the law of the venous system, with the **choroid plexuses** for its production and the **Pacchionian granulations** for its resorption. Lymph, let us remember, also returns to this system, participating in the overall balance.

THE CAPILLARY BEDS: MEETING AND TRANSFORMATION POINTS

The great plan of balance lies in the **capillary beds**, the meeting place between the arterial and venous systems. At this level, there is a transformation of energy, a rebalancing of rhythmic zones, and an adjustment of the **vasoregulation** of the different planes. These capillary beds are the "guardians of homeostatic balance" on the fluid plane, corresponding to areas of copula, specific rhythmic zones, essential for bodily balance.